

Collective dynamics and pathwise irreversibility in nonreciprocal networks of decentralized LLM agents

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Abstract

We define and measure a random-sequential stochastic process whose microscopic degrees of freedom are choices made by independent large-language-model (LLM) agents. Agent i owns a categorical belief $b_i \in \{-1, +1\}$, categorical action $a_i \in \{-1, +1\}$, bounded confidence and commitment, private field, bounded memory, inbox, outbox, and local workload. At each attempted update the scheduler exposes only the recorded local state and delivered packets; the LLM, not the scheduler, selects the next belief, action, signal, commitment, memory state, and typed tool. Reciprocal and directed delivery matrices share edge support, weighted in/out degree, opportunity schedules, packet schema, and inference seeds. We characterize isolated-agent kinetic response, collective order and fluctuations, projected probability currents, finite-sample-corrected block time-reversal divergence, and a symmetric-layer reference energy. The exact heat-bath model from the preceding theory study supplies an equilibrium and quadratic-onset reference, while a fitted kinetic-Ising process separates effective binary dynamics from residual LLM behavior. Neighbor state shifted individual choices by 0.0833 per unit field (95% CI 0.0417–0.1250), and coupling/noise changed finite-size order, susceptibility, and relaxation. By contrast, strong nonreciprocity produced no detectable excess path irreversibility in either small (-0.00023 , 95% CI -0.00144 – -0.00101) or collective (-0.00076 , -0.01115 – -0.00972) networks, and no replicated quadratic onset. The LLM process is not assumed Bayesian, and whenever first-order closure fails we interpret reversal divergence only as coarse-grained path irreversibility, not exact thermodynamic entropy production.

Keywords: nonequilibrium statistical mechanics; stochastic thermodynamics; nonreciprocal networks; agent-based systems.

1 Introduction

Statistical mechanics turns local stochastic rules into predictions about collective order, fluctuations, relaxation, and irreversible currents. Glauber dynamics provides a canonical equilibrium construction for binary variables [6]; Schnakenberg network theory and stochastic thermodynamics extend that language to finite Markov systems with sustained probability currents [9, 13, 15, 16]. The same formal ingredients occur in decentralized agent systems: local stochastic choice, neighborhood coupling, persistence, private fields, and directed information flow. What is usually missing is an empirical microscopic process whose states are actually chosen by autonomous agents rather than assigned by an external spin simulator.

Large language models offer a stringent realization. Their choices are stochastic and context dependent; their hidden activations, sampling state, and linguistic history need not close under a small visible state. They can also be instantiated as genuinely separate agents with private observations and explicit message delivery. This makes them scientifically interesting and methodologically dangerous. A binary label can support useful kinetic observables without being a physical spin, but a pathwise KL divergence cannot be called exact entropy production unless the observed process has an adequate stationary Markov representation. Hidden-variable and coarse-graining results make this distinction essential [8, 12, 17].

Recent work maps LLM populations to alignment or conformity models and reports topology-dependent finite-size collective behavior [3, 7, 10]. Those studies establish a useful connection to statistical physics but do not isolate degree- and traffic-matched nonreciprocity while measuring state-space currents. Conversely, asymmetric kinetic-Ising theory directly relates directed coupling to irreversibility [1, 4, 19], but its binary units are prescribed stochastic variables rather than text-conditioned agents with private memory, messages, commitments, and typed local consequences.

Here we connect these levels prospectively. We first fit an individual kinetic response law from controlled information states. We then run the same pinned LLM as independent agents on reciprocal and nonreciprocal fixed-degree networks. The design counterbalances semantic labels, holds communication quantity fixed, uses paired orientations and exogenous stochastic tapes, and treats a whole graph trajectory as the independent unit. Our claims concern the measured choice process, not intelligence, application performance, physical heat, or a thermodynamic limit.

2 Related statistical mechanics and agent literature

Logit response in strategic interaction already admits a statistical-mechanical interpretation when utilities derive from a potential [2]. Reciprocal binary couplings similarly admit a Gibbs reference; asymmetric couplings generically break detailed balance. Modern treatments of asymmetric kinetic-Ising systems use response, delayed correlations, and entropy production to characterize the resulting dynamics [1]. Nonreciprocal interactions can also create collective states with no equilibrium analogue [5]; our finite LLM networks are too small to establish such a phase transition, so we report finite-size collective-regime changes only.

Trajectory reversal provides an observable route to irreversibility. For a stationary process, the KL divergence between forward and reverse paths lower-bounds dissipative entropy production when relevant hidden coordinates are discarded [8, 12]. Transfer entropy instead measures lagged predictive information and is not itself entropy production [14]. Both quantities suffer finite-sample bias, particularly for sparse categorical words; we therefore report pseudocount sensitivity, time-shuffled floors, information-permutation floors, and graph-level intervals [11].

The closest LLM studies probe conformity, intrinsic field bias, and temperature-driven crossovers [3, 7, 10]. V12 adds separate belief and executed-action layers, directed typed packets, degree-preserving orientation pairs, state-current estimators, and an explicit comparison among exact heat-bath theory, a fitted binary surrogate, and direct LLM paths. The contribution is not the general fact that asymmetry can create entropy production; it is the controlled empirical realization and the measurement boundary imposed by LLM memory and coarse graining.

Scheduler offers one random-sequential update; it never selects the response

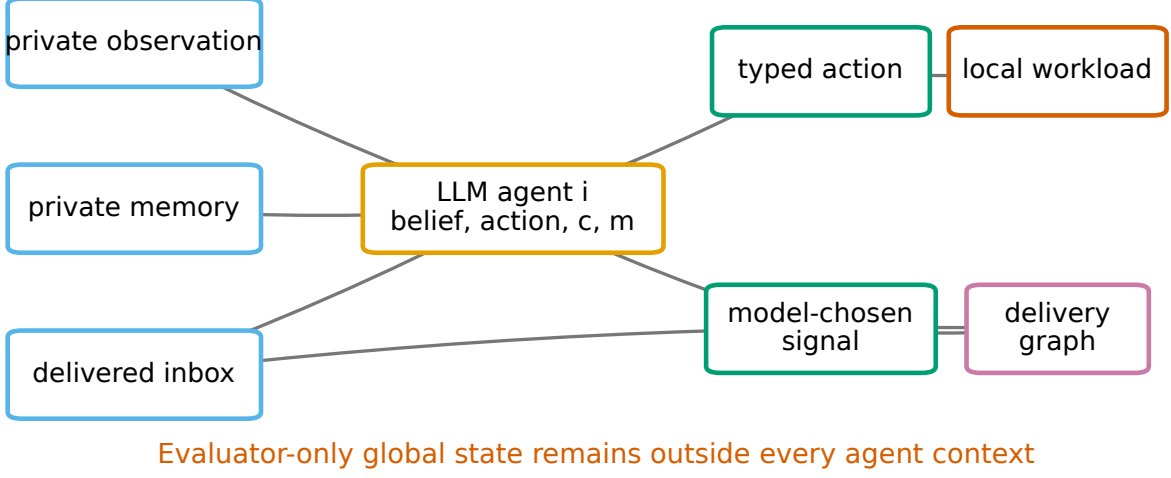


Figure 1: Independent-agent information and authority boundary. A scheduler offers a random-sequential update and transports a validated model-selected packet; it never selects the belief, action, signal, commitment, memory state, or tool. Global state is evaluator-only.

3 Decentralized LLM-agent microdynamics

3.1 Agent state and information boundary

For agent i at attempted update t , the recorded local state is

$$x_i(t) = (b_i(t), a_i(t), c_i(t), q_i(t), m_i(t), w_i(t)), \quad (1)$$

where $b_i, a_i \in \{-1, +1\}$ are decoded categorical belief and action, $c_i \in [0, 1]$ is self-reported confidence, q_i is one of five bounded commitment states, m_i is one of five explicit memory states, and w_i is a bounded local workload. Each agent additionally owns a fixed private field $h_i \in \{-1, 0, +1\}$, a role, inbox, outbox, and—only in the secondary regime—a three-entry private memory history. The evaluator records $X_t = (x_1, \dots, x_N)$ and message state, but no agent receives X_t .

The LLM returns exactly one schema-validated object containing categorical belief and action, confidence, commitment, bounded memory state, outgoing signal, typed tool, and reason code. A failed response receives one deterministic format repair. If that also fails, the attempted update leaves the scientific state unchanged; no centralized replacement is inserted. Executing the selected tool changes only the scheduled agent’s bounded workload and can improve or worsen it depending on the relation between the LLM-selected action and private field. This creates a consequential local action without turning an application simulator into the primary contribution.

3.2 Markovized and persistent-memory regimes

The primary prompt is rebuilt at each attempt from the recorded current local state, delivered inbox, private field, role, protocol text, and current clock. No chat history is retained. This is a Markovization intervention, not proof of first-order closure: model sampling and omitted internal variables may still leave detectable history dependence. The

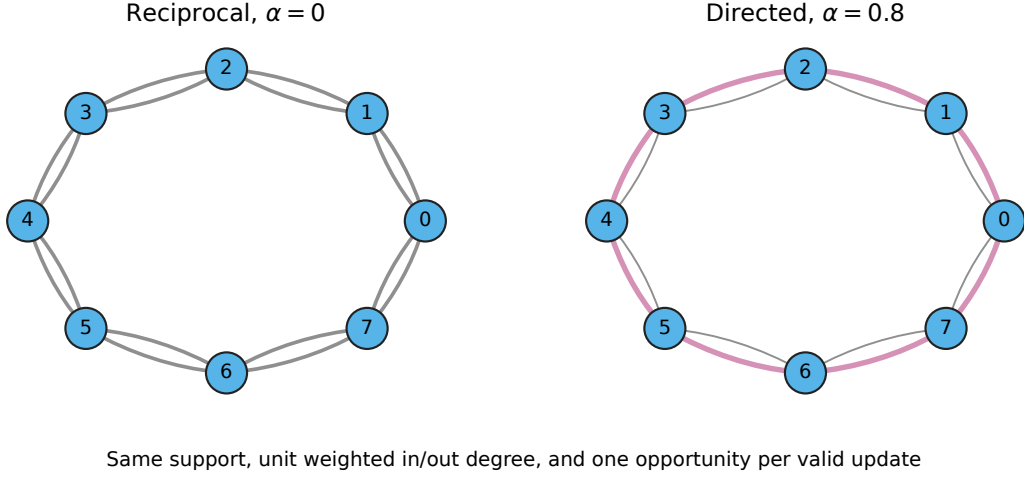


Figure 2: Reciprocal and directed matrices on common support. Directionality changes recipient probabilities without changing weighted degree or message opportunities.

secondary persistent-memory regime adds only the agent’s own last three bounded state summaries. We compare history-order likelihoods and conditional mutual information in both regimes.

The pinned model is Qwen2.5-7B-Instruct [18], revision `a09a35458c702b33eeacc393d103063234e8bc28`, loaded with NF4 double quantization and BF16 computation. Sampling temperature and top- p are generation controls. We do not identify decoding temperature with physical or effective thermodynamic temperature.

3.3 Latent-label neutralization

The preceding qualification study found a strong displayed-right preference. V12 treats this as an external-field artifact. Amber and cobalt are described symmetrically; which label decodes to latent +1 is counterbalanced, display order is randomized, private fields and disordered initial states are balanced, and the same mapping and random tape are used within matched network arms. Only decoded spins enter statistical analysis.

4 Reciprocal and nonreciprocal delivery networks

Let A be a symmetric fixed-degree support and W_s its symmetric row-stochastic weight matrix. A cycle-basis construction gives an antisymmetric, divergence-free circulation A_a , with $A_a^\top = -A_a$ and $A_a \mathbf{1} = 0$. Delivery weights are

$$W(\alpha) = W_s + \alpha A_a, \quad \alpha \in \{0, 0.20, 0.50, 0.80\}. \quad (2)$$

The circulation is normalized so every weight remains nonnegative at the largest α . Thus support, total weight, every weighted in/out degree, one message opportunity per valid update, packet schema, and payload size remain fixed. $W(\alpha)^\top$ supplies the paired reverse orientation.

We use a degree-two ring and a degree-three modular graph. Keeping degree fixed is important: the earlier V10 modular construction increased degree with N , saturated an unnormalized local field, and produced apparent large- N freezing. That historical anomaly is not used as V12 evidence.

At each attempted update, the precomputed tape selects one agent, display-order paraphrase, sampling seed, and recipient uniform variate. The LLM independently selects packet content. The delivery matrix maps the common recipient variate to an authorized neighbor. One sweep is N attempted updates.

5 Equilibrium-reference observable and empirical response models

Using the symmetric influence layer, we record the effective reference energy

$$H_{\text{ref}}(X) = -\frac{J_b}{2}\mathbf{b}^\top W_s \mathbf{b} - \frac{J_a}{2}\mathbf{a}^\top D_s \mathbf{a} - K\mathbf{a}^\top \mathbf{b} - \mathbf{h}^\top \mathbf{b} - \mathbf{g}^\top \mathbf{a}. \quad (3)$$

Here $D_s = W_s$, $(J_b, J_a, K) = (1, 0.65, 0.8)$ define a dimensionless diagnostic, and $\mathbf{g} = 0$ in the primary network. This is not a Hamiltonian claimed to generate Qwen’s stationary distribution. In the V10 heat-bath reference the same form does define a Gibbs process under reciprocal coupling; its exact detailed-balance and quadratic-onset results are retained solely as a mechanistic reference.

Isolated-agent calls span private field, delivered neighbor field, current belief/action, coupling relevance, decoding-noise control, and memory regime. We fit held-out logistic kinetic, nonlinear additive, and persistence-interaction models for both belief and action. A separate fitted kinetic-Ising surrogate is then simulated under the V12 graphs. It receives no LLM trajectory outcome and is labeled an effective comparator rather than LLM evidence.

6 Statistical-mechanics observables

The belief and action order parameters and cross-layer overlap are

$$m_b = N^{-1} \sum_i b_i, \quad m_a = N^{-1} \sum_i a_i, \quad q_{ba} = N^{-1} \sum_i b_i a_i. \quad (4)$$

We record edge disagreement density, $N \text{Var}(m_b)$ and $N \text{Var}(m_a)$ without temperature normalization, graph-distance correlations, integrated autocorrelation time, and relaxation from ordered and disordered starts. A predeclared field sweep supplies finite-system hysteresis area; no thermodynamic-limit hysteresis is inferred.

For $N \leq 4$, the primary categorical projection is the exact integer encoding of $(\mathbf{b}, \mathbf{a})$. For larger N , a fixed coarse macrostate bins $(m_b, m_a, q_{ba}, H_{\text{ref}}/N)$. We estimate Shannon configuration and macrostate entropy with Miller–Madow sensitivity, and report Tsallis $q \in \{0.5, 2\}$ and Gini–Simpson diversity only as predeclared secondary summaries. There is no search over q .

Neighbor mutual information and directed transfer entropy use permutation floors. The latter is a predictive dependence statistic, not a thermodynamic rate. Layer-projected belief and action irreversibility are likewise diagnostic projections and are not assumed additive.

7 Currents, entropy production, and path reversal

For an adequately sampled first-order categorical path, transition counts give a regularized kernel P_{xy} and stationary vector π . The projected current and Markov entropy-production

statistic are

$$J_{xy} = \pi_x P_{xy} - \pi_y P_{yx}, \quad (5)$$

$$\hat{\sigma}_M = \frac{1}{2} \sum_{x,y} J_{xy} \log \frac{\pi_x P_{xy}}{\pi_y P_{yx}}. \quad (6)$$

We report nats per attempted update and $N\hat{\sigma}_M$ nats per sweep, with pseudocounts 0.1, 0.5, 1.0. Because sparse smoothing can itself produce a positive estimate, reciprocal values and shuffled controls are essential.

Our primary scalable irreversibility statistic is block reversal divergence. For a word length L ,

$$\hat{\mathcal{I}}_L = \frac{1}{L-1} \sum_{\omega} \hat{p}(\omega) \log \frac{\hat{p}(\omega)}{\hat{p}(\omega^R)}. \quad (7)$$

The primary $L = 3$ value is corrected by the mean of 200 within-panel time-shuffled paths. We also examine $L = 2, 4$, coarse-graining widths, burn-in, and trajectory truncation. Excess irreversibility subtracts the matched reciprocal panel,

$$\Delta\mathcal{I}_L(\alpha) = \mathcal{I}_L(\alpha) - \mathcal{I}_L(0). \quad (8)$$

When Markov closure is inadequate, Eq. (7) is described only as coarse-grained pathwise irreversibility or a lower bound on hidden dissipation.

8 Prospective experimental design

An engineering pilot could inspect only schema validity, latent occupancy, bidirectional transition counts, delivery, privacy, and runtime. It used only reciprocal graphs and never computed currents, reversal divergence, or a nonreciprocity effect. The final attempt had balanced occupancy and passed all estimability criteria. The formal protocol was then frozen at SHA-256 `cee49ca7111b81ae9544a8d32a754ce4c9f15bff16333e98588147a7f7f665b6` with execution source `0796286362ec4dde0eb4f2dc88ecea4c3bf53859e618aec1928ba8b9e8b0a154`.

The isolated response block contains 576 decisions. Small systems use $N \in \{3, 4\}$, four graph/environment replications, 60 sweeps, and nine-sweep burn-in. Collective systems use $N \in \{8, 16\}$, ring and modular graphs, a 2×2 grid of neighbor relevance (0.35, 0.80) and decoding-noise control (0.50, 0.85), two replications, eight sweeps, and two-sweep burn-in. Exact matched blocks cover bounded memory, relaxation, six communication controls, and a bidirectional field sweep. Every reciprocal arm is paired with forward and transposed orientations at positive α .

Graph trajectory is the independent unit. Primary intervals use 10,000 cluster bootstrap resamples; secondary families use Holm correction. Quadratic, linear, and linear-plus-quadratic weak- α models are compared by leave-cluster-out prediction and descriptive AICc. No result-dependent stop or parameter change is allowed after freeze.

The first post-formal analysis invocation stopped before writing result tables because a replicate identifier present in the frozen design was not copied into the derived panel-summary row used by the factor analysis. A separately logged bookkeeping wrapper copied that existing identifier and reran the unchanged estimands. No trajectory, grouping rule, contrast, seed, threshold, or exclusion changed; the failed invocation and repair checksum are retained.

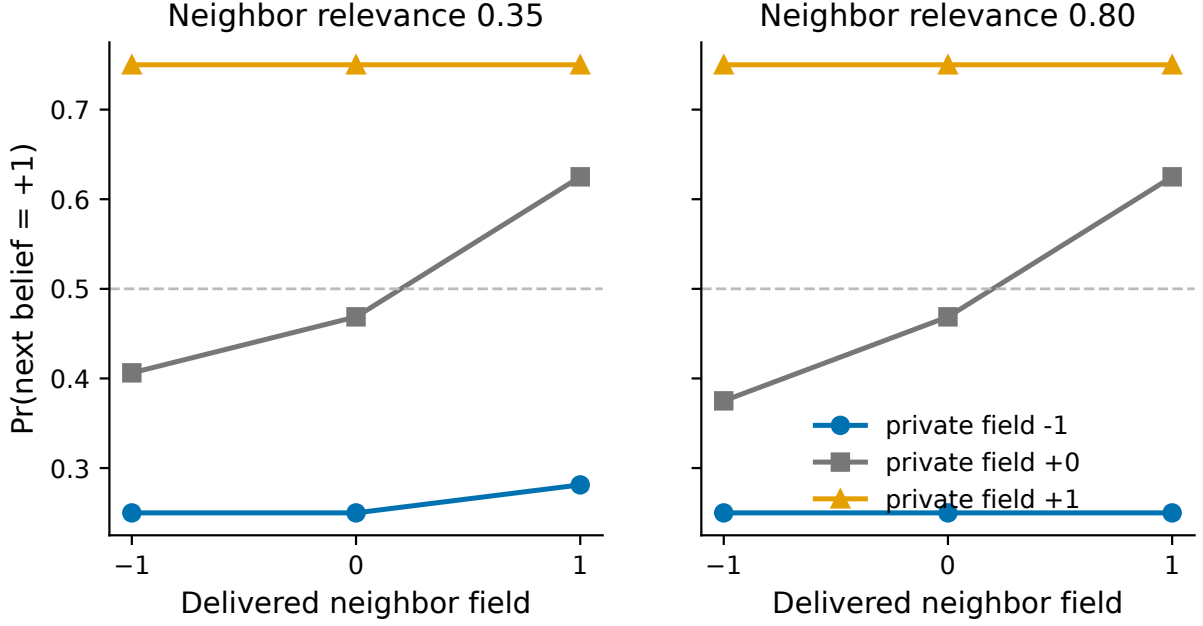


Figure 3: Categorical next-belief response to private and delivered-neighbor fields. The latent label is counterbalanced before aggregation.

9 Individual-agent response results

Delivered neighbor state changed the categorical LLM response after latent-label counterbalancing: the paired increase in latent-+1 choice per unit neighbor field was 0.0833 (cluster-bootstrap 95% CI [0.0417, 0.1250]; 96 independent information-state clusters). Both belief transition directions occurred 58 times and the isolated-agent occupancy was exactly 0.500. Thus H1 was supported.

The kinetic, nonlinear, and persistence-aware held-out losses quantify how much of the local response can be represented by an effective binary model. Fitted coefficients are response parameters, not a physical temperature unless the logit form and symmetry diagnostics support that interpretation.

10 Collective dynamics

Collective dynamics responded reproducibly to the microscopic controls. Increasing neighbor relevance from 0.35 to 0.80 increased $|m_b|$ by 0.0330 (95% CI [0.00861, 0.0591]), $N \text{Var}(m_b)$ by 0.0455 ([0.0228, 0.0677]), and integrated correlation time by 2.319 attempted updates ([0.781, 3.881]). Raising decoding-noise control from 0.50 to 0.85 decreased these quantities by 0.0539, 0.0736, and 5.990, respectively; all six effects survived Holm correction. H5 was supported as a finite-size collective-regime response, not a phase transition.

11 Probability currents and irreversibility

Strong nonreciprocity did not increase the primary bias-adjusted length-three path-reversal divergence. Directed-minus-reciprocal differences were -2.33×10^{-4} nats/update (95% CI $[-1.44 \times 10^{-3}, 1.01 \times 10^{-3}]$; 8 clusters) for $N = 3, 4$, and -7.59×10^{-4} (95% CI

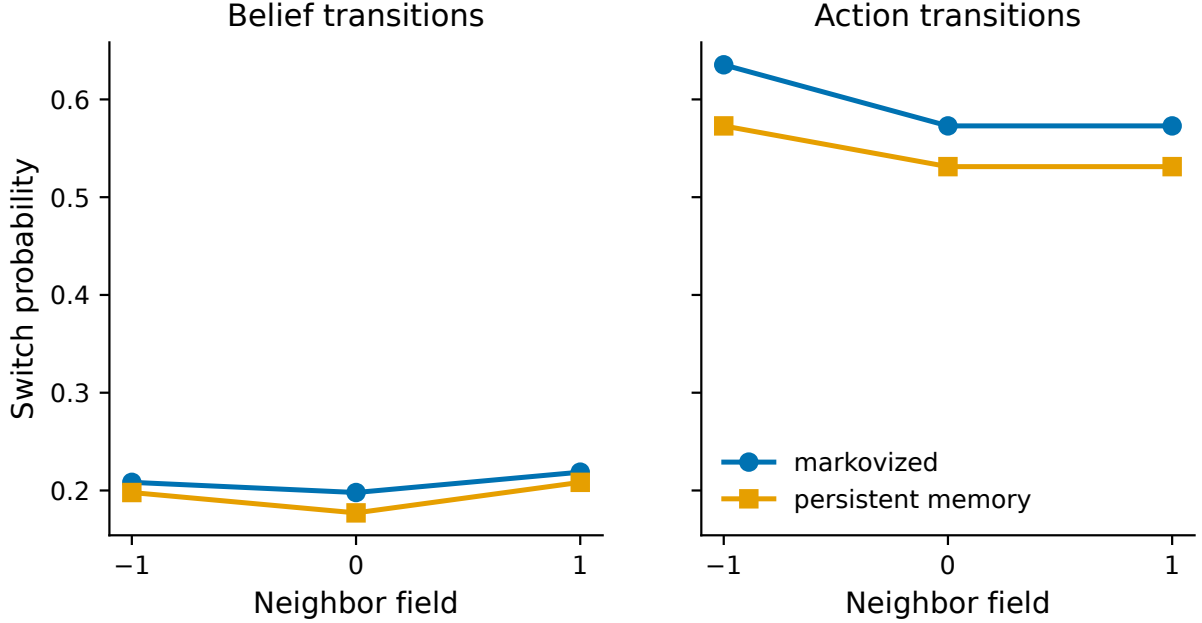


Figure 4: Belief and action switching under Markovized and bounded-memory prompts. Switching is measured from realized choices, not self-reported probability.

$[-1.12 \times 10^{-2}, 9.72 \times 10^{-3}]$; 32 clusters) for $N = 8, 16$. H2 was not supported. The predeclared dose slopes were also null: 7.07×10^{-4} (95% CI $[-5.01 \times 10^{-4}, 2.33 \times 10^{-3}]$) in small networks and -5.39×10^{-5} (95% CI $[-1.45 \times 10^{-2}, 1.46 \times 10^{-2}]$) in collective networks. Only 25% and 15.6% of clusters, respectively, were monotone. H3 was not supported. In small networks the constrained quadratic coefficient was positive, 0.00604 (95% CI $[7.48 \times 10^{-5}, 0.0140]$), but its leave-cluster-out RMSE (0.001937) was slightly worse than the linear model (0.001925). In collective networks the quadratic coefficient was 0.00547 with a wide interval $[-0.0353, 0.0481]$. The required positive, predictively preferred quadratic component was therefore absent and H4 was not supported. Direction and size did not replicate. At $N = 8$ the two strong- α orientation estimates were positive but imprecise (0.00444 and 0.00769 nats/update); at $N = 16$ both were negative (-0.00756 and -0.00761), with every interval crossing zero. Small-system orientation effects also changed sign across $N = 3, 4$. H6 was not supported.

12 Size, memory, and controls

First-order closure was adequate only as a working projection. For the small-network categorical path, median conditional mutual information beyond one step was 4.4×10^{-5} nats and median early-late Jensen-Shannon drift was 0.0034. The collective macrostate path had corresponding medians 0.00785 and 0.348, revealing material history dependence and nonstationarity over the short eight-sweep trajectories. Accordingly, projected Markov currents are descriptive diagnostics; block reversal is the primary coarse-grained irreversibility measure. Adding explicit private memory increased adjusted block reversal by 0.0179 nats/update (95% CI $[0.00357, 0.0340]$) but did not make a total-entropy-production interpretation exact. No communication, content/time/sender permutation, and placebo controls each had only three independent clusters. None survived Holm correction. The no-message arm had 0.0409 higher adjusted reversal than unaltered messaging (95% CI

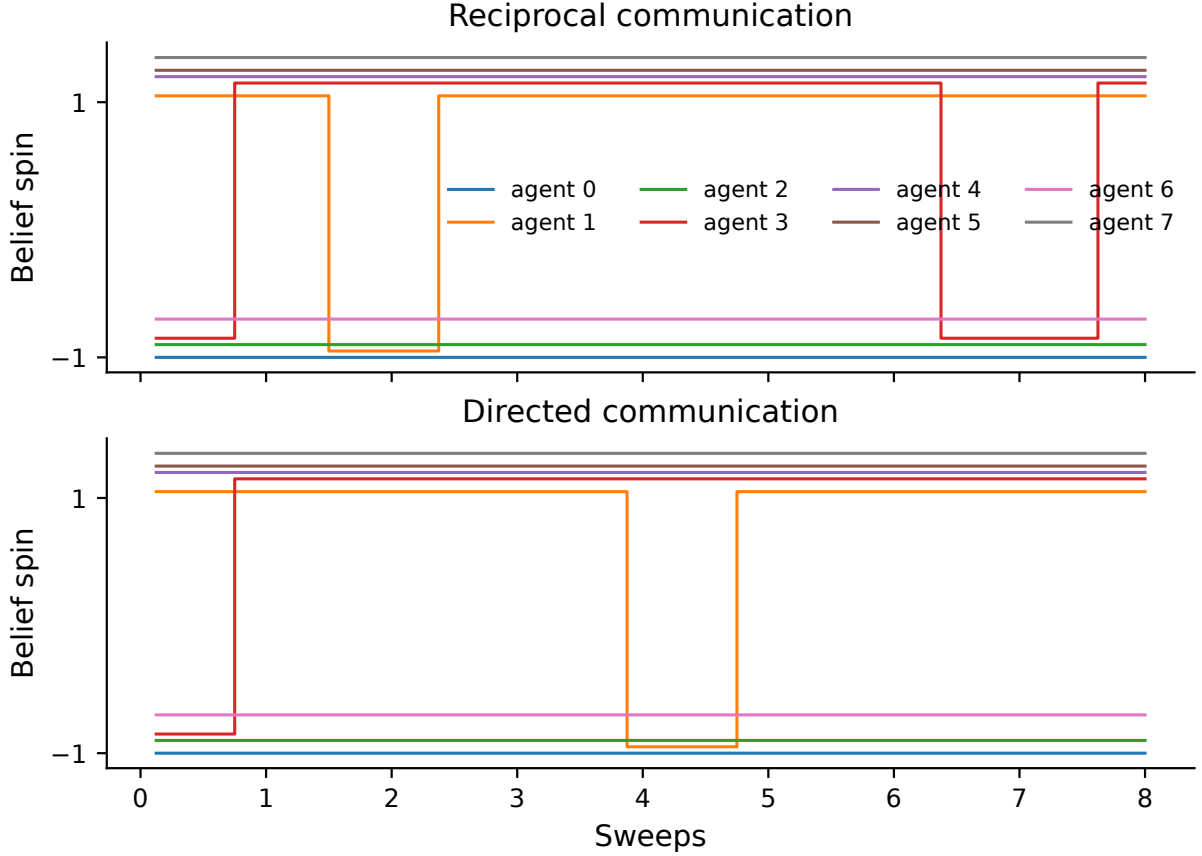


Figure 5: Predeclared matched reciprocal and directed network trajectories. Offsets distinguish persistent agent identities.

[0.00331, 0.0760]), but its paired sign-flip $p = 0.249$ and Holm-adjusted $p = 1$ expose the low control-arm resolution. These controls therefore constrain interpretation without rescuing the nonreciprocity hypothesis.

13 Discussion

V12 separates three evidentiary levels. The heat-bath reference has an exact finite-state Gibbs equilibrium in its reciprocal limit and an exact finite-kernel entropy-production rate. The fitted kinetic-Ising process is a numerical effective model learned from isolated categorical responses. The LLM results are empirical trajectories generated by independent contexts and local messages. Agreement among their curve shapes is mechanistic evidence; it does not make their numerical rates identical.

The design also separates autonomy from anthropomorphism. Each LLM call owns and changes a persistent local state, chooses a consequential local tool, and emits content delivered only through authorized edges. No global optimizer chooses the responses. Nevertheless, these are synthetic binary coordination choices from one instruction model, not human organizations or field-deployed agents.

Potential nonzero reciprocal irreversibility has several interpretations: finite-sample bias, option or prompt asymmetry, cyclic scan structure, hidden model state, irreversible commitment/workload coordinates, or a genuinely non-potential local response. Subtracting the paired reciprocal arm isolates an excess associated with directed delivery but does not

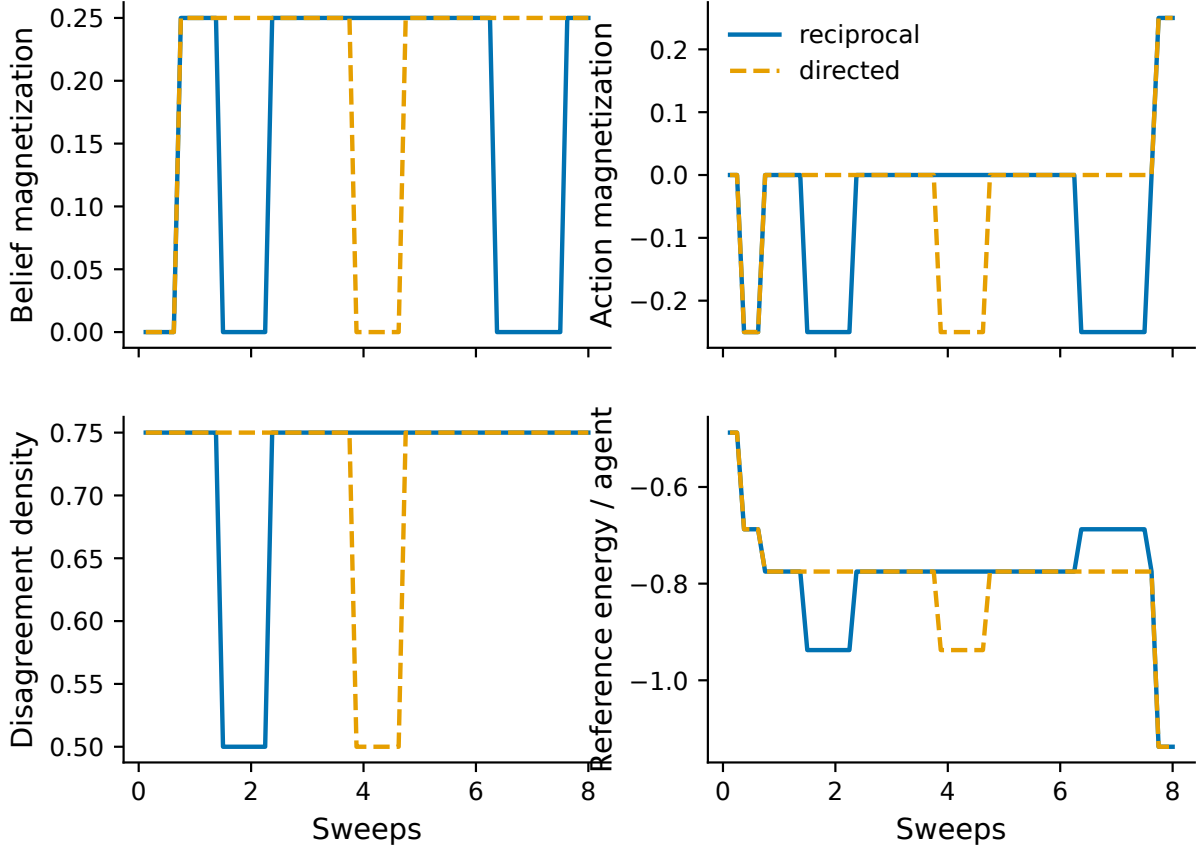


Figure 6: Belief/action order, edge disagreement, and reference energy along matched representative trajectories.

identify literal heat. Message permutation and placebo controls ask whether the excess depends on informative directed content rather than prompt volume alone.

The principal empirical result is a boundary rather than the proposed positive nonreciprocity effect. The LLM’s microscopic transition law was measurably neighbor-sensitive, and ordinary coupling and decoding-noise controls shifted collective order, fluctuations, and relaxation. Degree- and traffic-matched directionality nevertheless produced neither a replicated excess reversal divergence nor the V10 heat-bath model’s quadratic onset. This separates the existence of local social response from the stronger claim that an asymmetric delivery graph creates an observable irreversible current on the sampled time and state scales.

The same construction can be interpreted as private reports passed among humanitarian coordinators or one-way telemetry among defensive utility agents. Those mappings change the semantics of fields, commitments, and messages but not the present estimand. We did not simulate service delivery, infrastructure restoration, or human decisions in V12 and make no application-performance claim.

14 Limitations

First, the primary model is one quantized Qwen revision. Inference seeds and graph realizations support replication within that implementation, not model universality. Second, the large-network paths are short because every attempted update is a GPU generation.

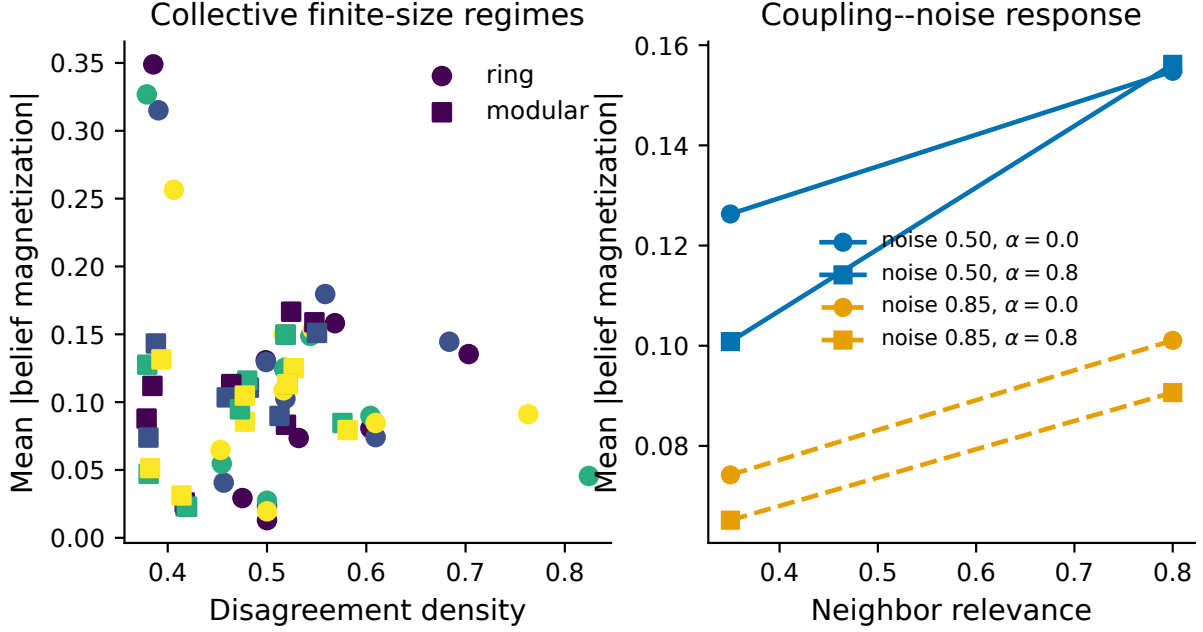


Figure 7: Finite-size collective regimes over coupling relevance and decoding noise. These panels are not claimed as a thermodynamic phase diagram.

Their collective summaries are finite-time and finite-size. Third, categorical coarse graining hides token-level and activation dynamics. Block reversal divergence is therefore a lower-bound-style observable when closure fails. Fourth, the scheduler’s update opportunity and a recipient uniform are exogenous; the primary experiment intentionally fixes opportunity volume to identify topology. Fifth, the typed workload is a bounded abstract consequence. Humanitarian coordination and defensive utility restoration are interpretive mappings only; no application benefit is tested.

15 Conclusion

We provide a reproducible stochastic-mechanical construction in which actual decentralized LLM agents, rather than a central spin simulator, generate the microscopic transitions. Fixed-degree reciprocal/directed pairs permit a clean test of information-flow asymmetry; local response models connect language decisions to kinetic binary dynamics; current and reversal estimators state their normalization and bias floor. The resulting claim is deliberately conditional on the observed data and Markov diagnostics. Negative, mixed, and positive hypotheses are retained alike.

For this pinned model, the supported claim is that independent LLM choices form a coupled, fluctuating finite network whose order and relaxation respond to neighbor information and decision noise. The stronger claim that controlled nonreciprocity generates measurable excess irreversibility is not supported at the sampled sizes and trajectory lengths. That distinction is the main statistical-mechanical result.

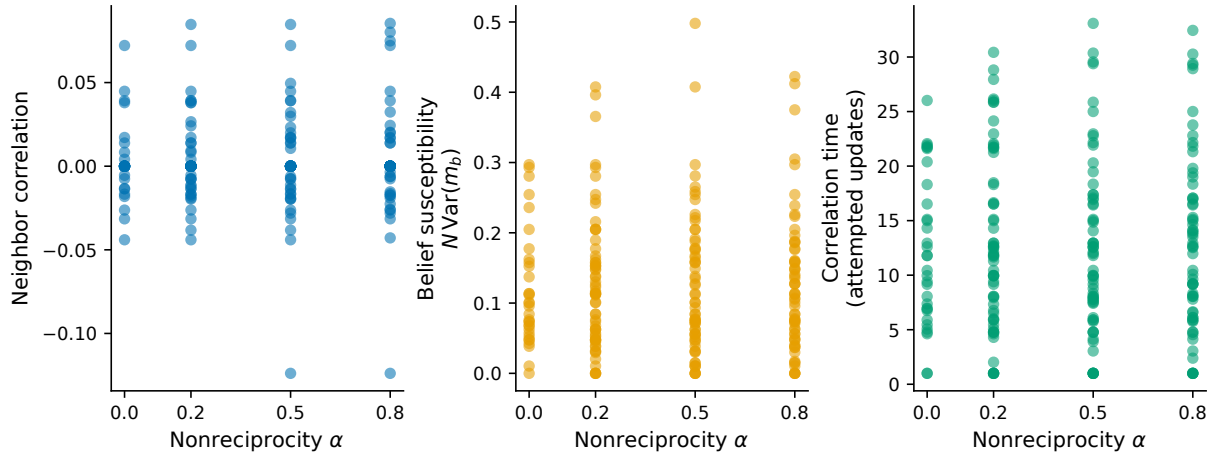


Figure 8: Correlation, unnormalized susceptibility, and integrated correlation time across directed-coupling conditions. Points retain graph-panel variation.

Reproducibility and data availability

The formal design contains 44,352 attempted LLM decisions and 44,882 model calls, with 24,509,789 prompt and 3,585,830 generated tokens. Metered generation time was 20.318 GPU-hours. Raw prompts, completions, and per-transition ledgers remain outside Git under `/workspace/ThermoAgent-v12-artifacts`; compact tree checksums, protocol, aggregate tables, figure source data, and exact replay summary are included with the repository-facing package. No chain of thought is requested or stored. Every content-addressed decision can regenerate its state transition without another model call.

AI-assisted preparation disclosure. OpenAI Codex was used under researcher direction for software implementation, execution orchestration, analysis checks, and drafting of this working manuscript. All scientific decisions, source citations, numerical claims, and final text require author review; the language model is not an author.

A Structured transition and wire packet

Every valid packet contains schema version, sender, time, belief spin, action spin, outgoing signal, quantized confidence, commitment code, memory code, and CRC framing. Primary arms use exactly the same binary serializer. Attempted, transmitted, delivered, dropped, and complete wire-byte counts are recorded separately.

B Estimator sensitivity

The compact tables retain raw plug-in Markov statistics, three pseudocounts, block lengths two through four, 200 shuffled floors per panel, augmented-state block divergence, early/late distributional drift, history-order one through three predictive likelihood, and conditional mutual information. Information metrics use 100 within-panel permutations. Uncertainty never treats messages, tokens, edges, or attempted updates as independent replicates.

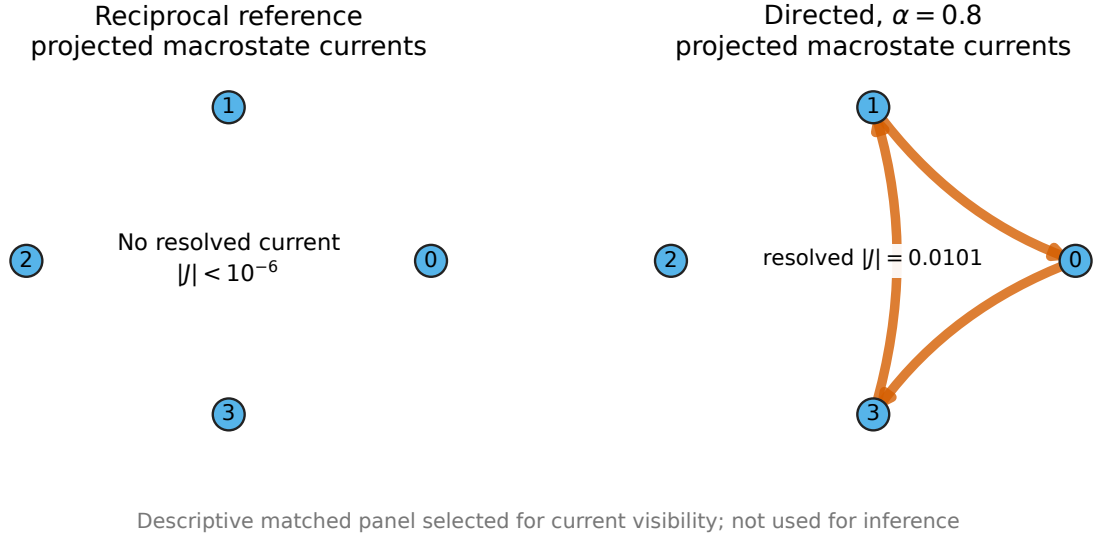


Figure 9: Projected currents in a descriptively selected reciprocal/directed matched macrostate pair. The pair was selected after analysis solely because it contained visible directed current and is not used for inference; the selection rule is retained with the figure source data.

C Prospective hypotheses

H1 tests whether delivered neighbor state changes the local LLM transition law after counterbalancing. H2 tests reciprocal versus directed current or path irreversibility at matched traffic. H3 tests a dose response to quantified nonreciprocity. H4 compares a positive quadratic weak- α component with linear alternatives and V10. H5 tests collective order/fluctuation response to coupling, noise, or directionality. H6 requires directional replication across paired orientations and two sizes. Scientific completion does not require all hypotheses to be positive.

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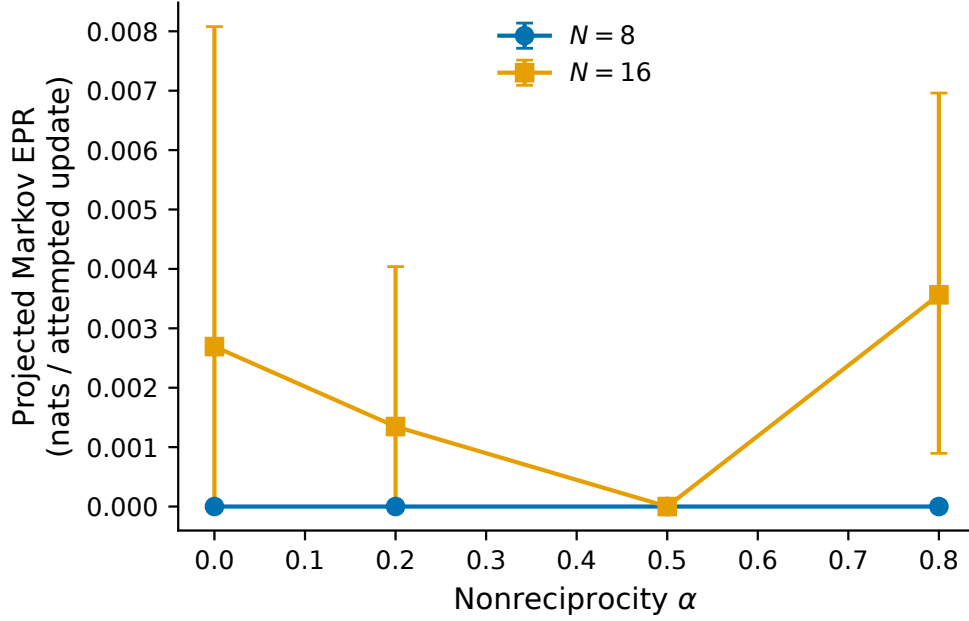


Figure 10: Regularized projected Markov entropy-production statistic. Its units are nats per attempted update; interpretation depends on closure diagnostics.

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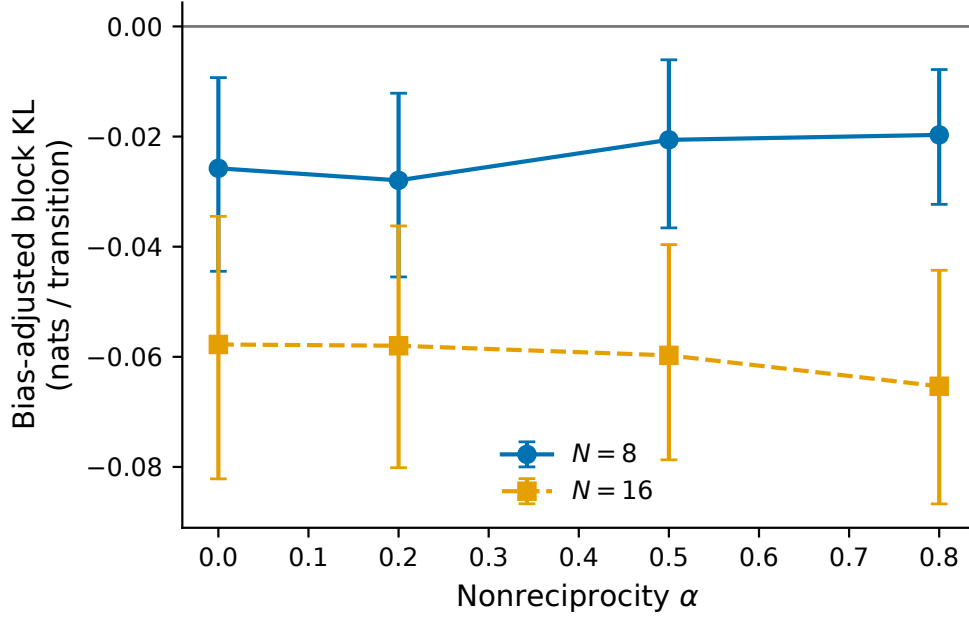


Figure 11: Bias-adjusted length-three block time-reversal divergence versus quantified nonreciprocity. Intervals resample independent matched clusters.

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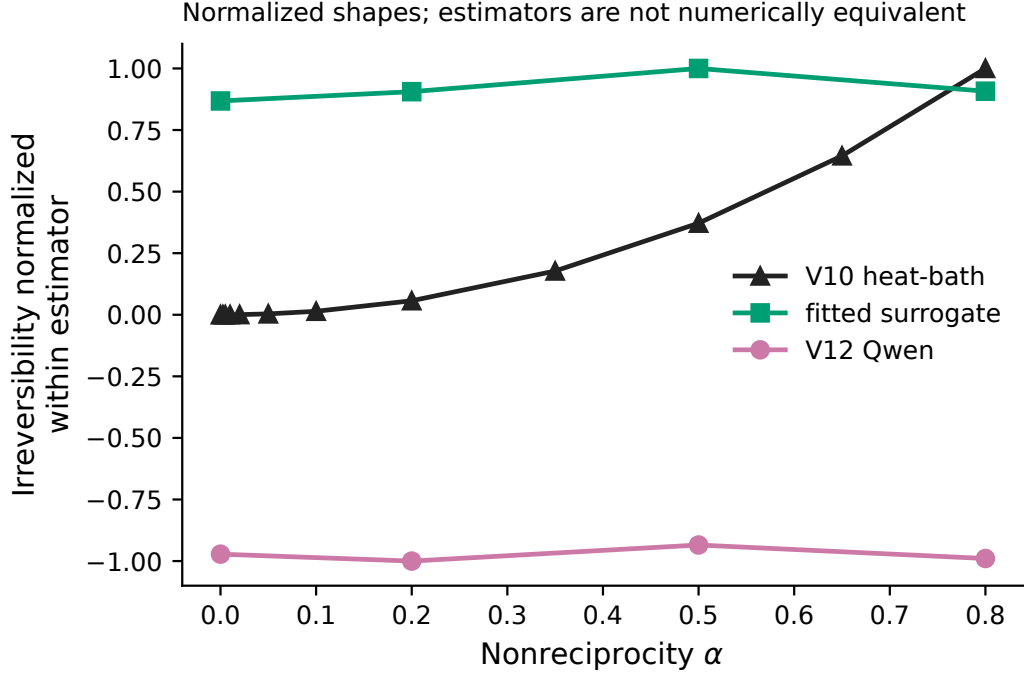


Figure 12: Normalized response shape for the exact V10 heat-bath model, fitted kinetic surrogate, and direct LLM trajectory. The three estimators are not numerically interchangeable.

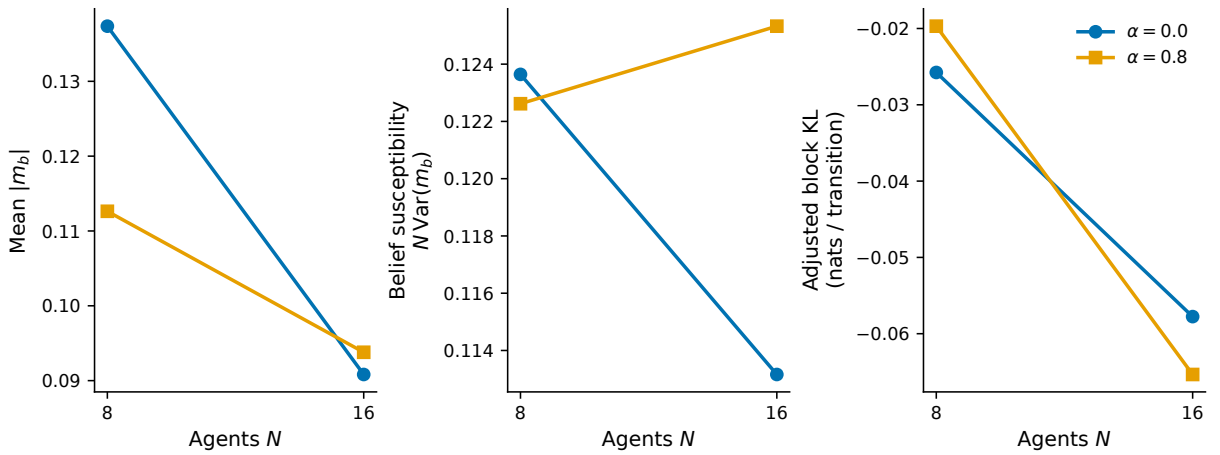


Figure 13: Order, fluctuations, and coarse path irreversibility at $N = 8, 16$. Two sizes can show size dependence but cannot establish thermodynamic scaling.

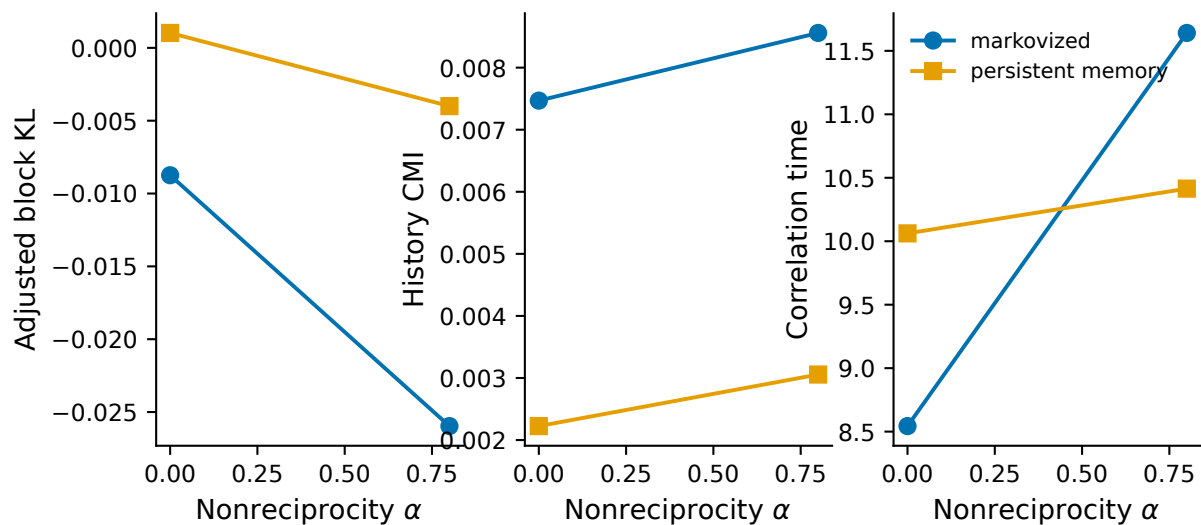


Figure 14: Exact matched comparison between rebuilt state-complete prompts and prompts containing a three-entry private memory.

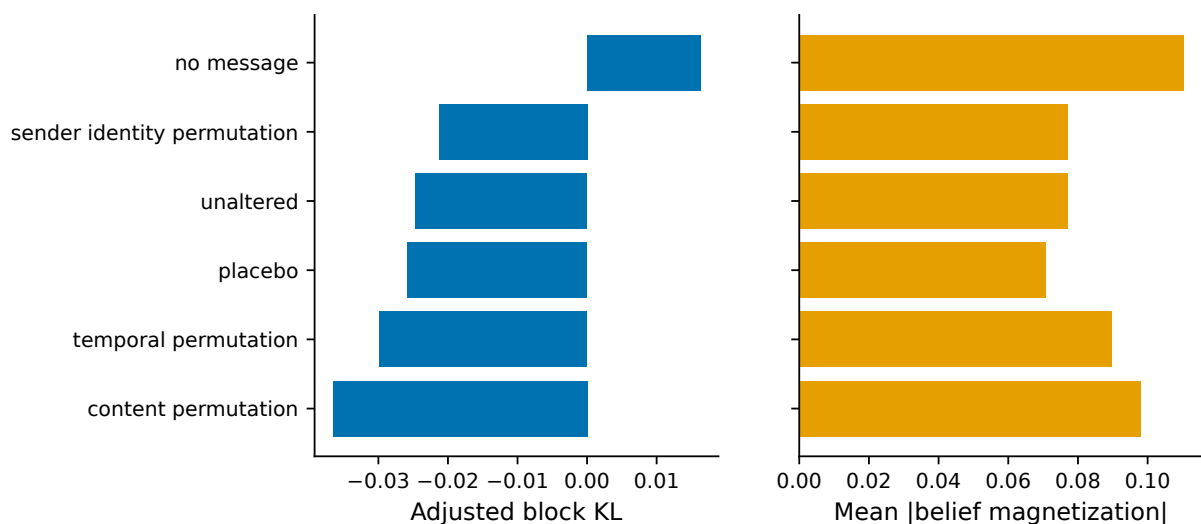


Figure 15: Message-content, time, sender, no-message, and placebo controls at fixed topology and opportunity count.

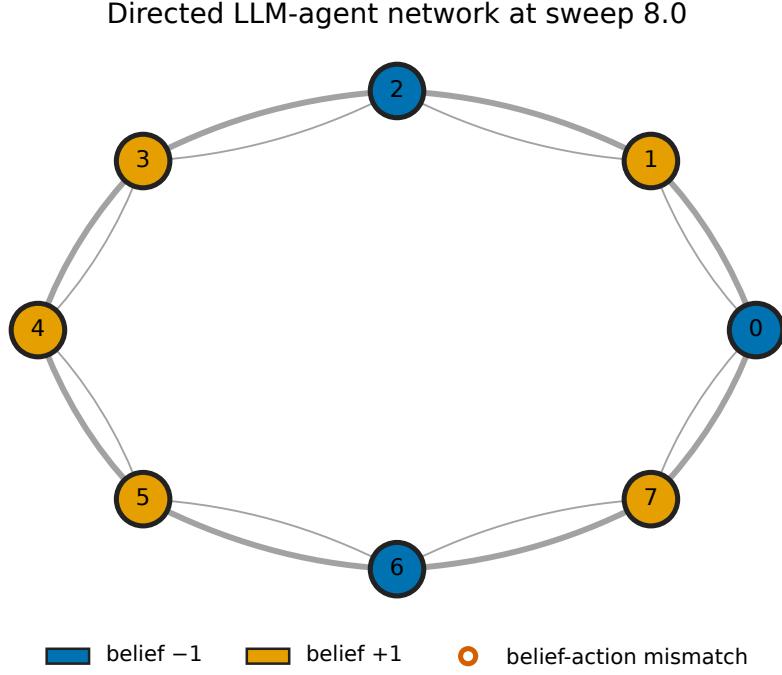


Figure 16: A directed-network snapshot. Fill denotes belief; outline denotes belief-action consistency; arrows denote supported delivery probabilities.

Table 1: Prospectively frozen primary hypotheses. Intervals resample the independent information-state cluster (H1) or matched graph-trajectory cluster (H2–H6).

Test	Primary estimate	Disposition
H1	Neighbor-field choice response 0.0833; 95% CI [0.0417, 0.1250]; $n = 96$ clusters	Supported
H2	Strong- α adjusted block-KL difference: small -0.00023 $[-0.00144, 0.00101]$; collective -0.00076 $[-0.01115, 0.00972]$	Not supported
H3	Dose slope: small 0.00071 $[-0.00050, 0.00233]$; collective -0.00005 $[-0.01449, 0.01456]$	Not supported
H4	Small-system quadratic term positive, but held-out RMSE exceeded the linear model; collective interval crossed zero	Not supported
H5	Coupling and decoding-noise controls changed order, susceptibility, and correlation time after Holm correction	Supported (finite size)
H6	Strong- α direction changed across sizes/orientations; all collective intervals crossed zero	Not supported